

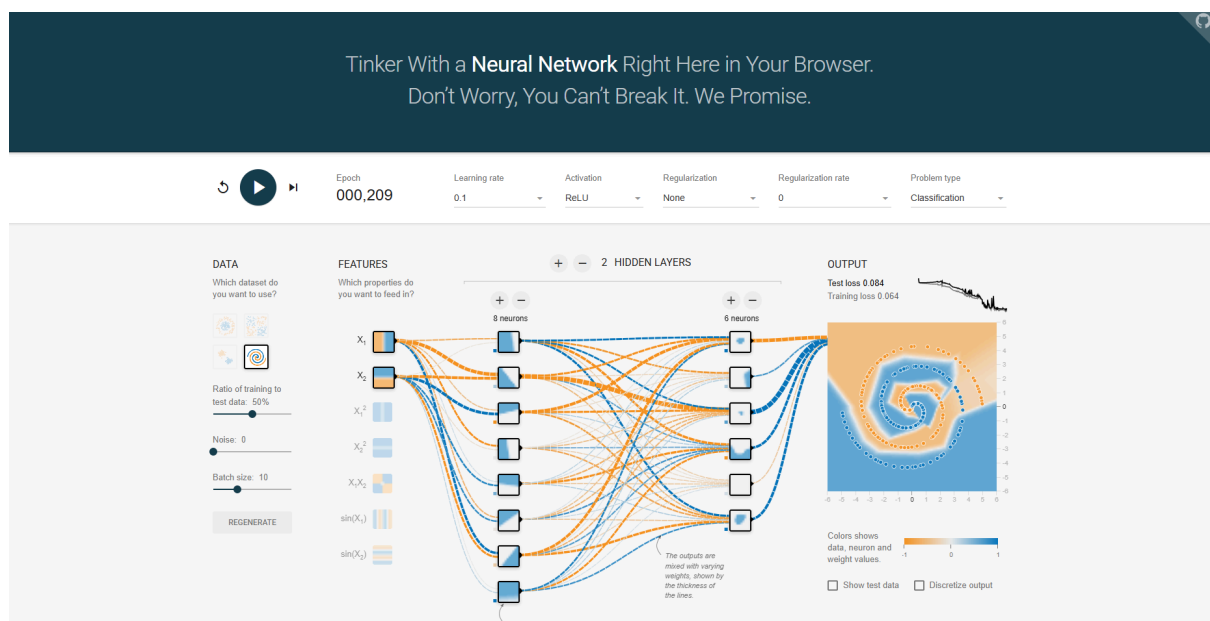
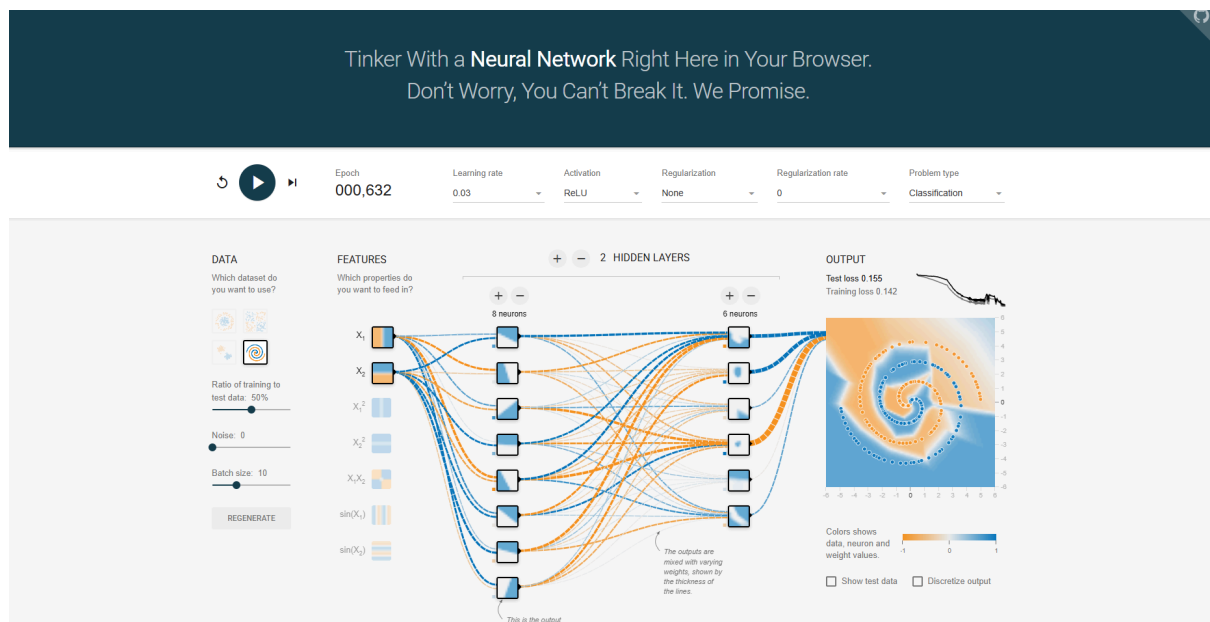
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Método de ativação utilizado: **ReLU**

Nos testes feitos, foi percebido os impactos das alterações de parâmetros na perda dos testes.

Com duas camadas, o aumento dos neurônios contribuiu para a diminuição da perda. Ao aumentar o learning rate, houve uma diminuição na quantidade de tempo (épocas) para atingir o valor de 0,1 de perda, porém ao aumentar demais foi vista uma instabilidade e não foi possível chegar no valor estabelecido de perda.

Ao aumentar a quantidade de camadas, foi possível uma diminuição da quantidade de neurônios em cada camada.



Tinker With a **Neural Network** Right Here in Your Browser.
Don't Worry, You Can't Break It. We Promise.

Epoch: 000,466
Learning rate: 0.3
Activation: ReLU
Regularization: None
Regularization rate: 0
Problem type: Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%

Noise: 0

Batch size: 10

REGENERATE

FEATURES

Which properties do you want to feed in?



X_1

X_2

X_1^2

X_2^2

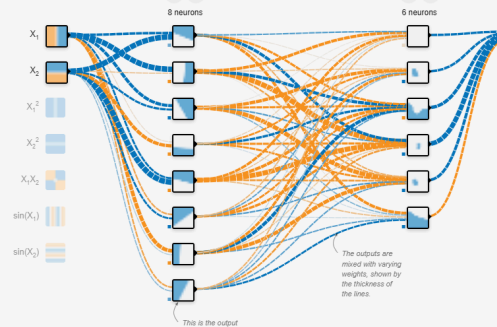
$X_1 X_2$

$\sin(X_1)$

$\sin(X_2)$

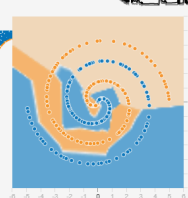
HIDDEN LAYERS

8 neurons 6 neurons



OUTPUT

Test loss 0.170
Training loss 0.140



Colors shows data, neuron and weight values.

☐ Show test data ☐ Discretize output

Tinker With a **Neural Network** Right Here in Your Browser.
Don't Worry, You Can't Break It. We Promise.

Epoch: 000,406
Learning rate: 0.1
Activation: ReLU
Regularization: None
Regularization rate: 0
Problem type: Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%

Noise: 0

Batch size: 10

REGENERATE

FEATURES

Which properties do you want to feed in?



X_1

X_2

X_1^2

X_2^2

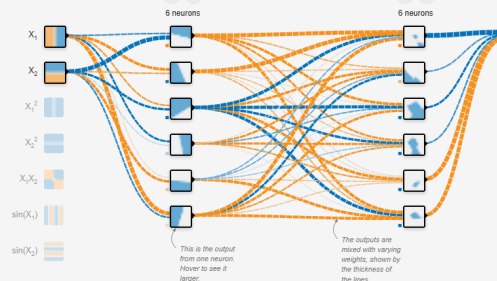
$X_1 X_2$

$\sin(X_1)$

$\sin(X_2)$

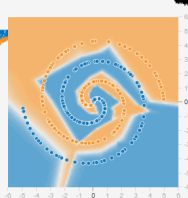
HIDDEN LAYERS

8 neurons 6 neurons



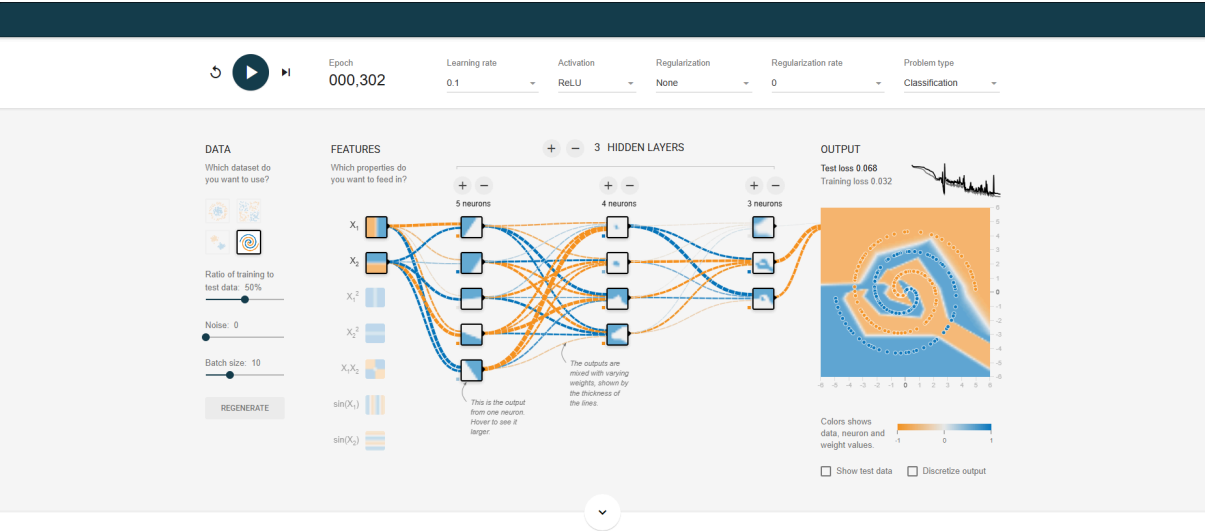
OUTPUT

Test loss 0.106
Training loss 0.081



Colors shows data, neuron and weight values.

☐ Show test data ☐ Discretize output



Um, What Is a Neural Network?

It's a technique for building a computer program that learns from data. It